

Brief for whoever defends this at the pre-screening

Self-contained. Assumes you have not read anything else. ~15 minutes.

THE 30-SECOND VERSION

If you remember nothing else, remember this:

Nagpur's water is tested. By three different government bodies. The results are published — as one city-wide percentage, as scattered points in a 295-page PDF, and as 37 separate map files.

So no resident of this city can answer "is my water safe?" about the place they actually live.

We didn't measure anything new. We took what the government already publishes and connected it to the person it's about.

Everything else is detail.

1. WHAT THIS PROJECT IS

A map of every published water measurement in Nagpur — 395 of them — where you can click any point, or any empty spot, and get a straight answer about what is known and what isn't.

And a rule we designed around: the system will never tell you your water is safe. It reports three states only:

- **Exceeds** a named legal limit — with the parameter, the value, the limit, and what to do
- **No exceedance** among the parameters actually tested there
- **Not tested** — no measurement exists for this location

Why we refuse to say "safe": certifying drinking water under IS 10500 requires bacteriological testing. We have that for only a handful of points. **Claiming safety we can't prove would be the one thing that discredits everything else.**

2. THE PROBLEM — with the actual numbers

Learn these. They're all from government sources.

Fact	Source
OCW publishes 99.6% of samples fit — one figure for 3 million people , no zone breakdown	OCW published data, to May 2026
CGWB sampled 347 groundwater points across the district. Median distance from the city centre: 35 km	CGWB aquifer mapping
Inside the city, NEERI sampled 12 groundwater sources. All 12 showed faecal coliform. IS 10500 requires zero	CSIR-NEERI for NMC, 2023-24
21 of 23 river sites fail CPCB Class D — the minimum for fish to survive	Same
Dissolved oxygen down the Nag: 3.67 → 0.48 mg/L. The collapse starts where a drain joins	Same
52 of 347 sampled sites exceed the nitrate limit	CGWB

The single best line:

"Nagpur has three million people. In the government's own 2023-24 report, twelve groundwater sources were sampled inside the city. Twelve. All twelve failed the bacteriological standard."

3. WHO THIS IS FOR — and what each person actually gets

Be specific when asked "who uses this." Vague answers lose.

The household on a borewell

Peri-urban and district households with no treatment at all. **Gets:** whether the nearest tested groundwater exceeded anything, and what to do about it. **Concretely:** if it's nitrate, we tell them boiling makes it *worse* — and that the risk is to bottle-fed infants under six months.

The farmer

Gets: whether the water is fit for irrigation under CPCB Class E, and what it costs a specific crop under FAO standards. Orange loses ~91% yield at the worst site we found; cotton loses nothing on the same water. **56 of 347 sites would cost an orange grower more than 10% yield. Zero would harm cotton.**

The city resident on municipal supply

Gets — honestly — mostly "we don't know about your tap." And that is the point, not a failure. What they do get: what's been measured near them, how far away it is, and the fact that a city-wide 99.6% cannot answer their question.

If asked "so it's useless for most people?" — see Q6 below. It has a good answer.

The person about to spend money

Drilling a borewell (₹1.5 lakh), buying an RO (₹18,000), renting a flat. **Gets:** what's known about water in that area before the money is spent.

The ward officer / municipal engineer

Gets: every agency's data on one screen. Currently it sits with a central board, a state board and a private operator, and no single one of them can produce this picture.

The journalist, researcher, or activist

Gets: a citable, attributed, dated view of the city's water — with the empty areas visible.

4. WHAT'S BUILT vs WHAT'S NEXT

Say this distinction out loud. It's what gets you believed on both halves.

Working now: - The map — 395 real measurements over 302 watercourses - Click any point → verdict, values, source agency, sampling period - Click anywhere → nearest actual measurement **and the distance to it** - What to do about each exceedance, including the boiling warning

Next: river oxygen profile · crop advisory · citizen-submitted readings · which sewage outfalls are upstream of you · supply hours by area · the flood layer.

Never say a planned feature is built. If asked, *"that one's next — here's what's running today."*

5. THE QUESTIONS — and how to answer

Q1 · "Doesn't this already exist?" (in any form)

"Yes. That's the point. Almost everything we show already exists — in a PDF, in a department, in a report nobody opens. None of it reaches the person it's about. We're not a new measurement. We're the delivery layer."

This one answer covers most versions of the question. Say it early.

Q2 · "Jal Jeevan Mission already publishes water quality."

"It does — village-level results, published publicly, texted to residents. **But JJM is rural only.** A farmer outside Nagpur can look up his water. A resident inside Nagpur cannot. We're asking why the city limit is where that stops."

Raise this yourself before a judge does. It looks like research; being caught by it looks like you didn't check.

Q3 · "Nagpur already has disaster management."

"It does, and it works — but **all of it is response.** It activates when water is already in the street. **NMC's own Monsoon Preparedness Plan lists 66 low-lying flood-prone areas.** The city already knows. That list has never reached the people living in those 66 areas. Disaster management asks *how do we respond?* We ask *why did nobody tell them?*"

Q4 · "The data is already public."

"It's a 295-page PDF, a 203-page report, and 37 separate map files. **Public and reachable are different things — and that gap is the entire product.**"

Q5 · "What's actually new?"

"Three things. **Connection** — nobody has joined a central board, a state board and a municipal operator's data, because they belong to three different bodies. **Resolution** — turning 99.6% city-wide into an answer about your location. And **honesty** — we refuse to say your water is safe, because we don't have the data to say it."

Q6 · "I know my water comes from NMC and it's filtered. Why do I need this?"

"NMC treats it at the plant. Your RO treats it at your tap. **Neither covers what happens in between** — and published research on

Nagpur found intermittent supply worsens microbial quality, because pipes that sit empty draw contamination in through leaks. Parts of this city get 18–24 hours of water; others get 2–4. **You have no way to know which you are.**

And honestly — **if you're on 24-hour supply with a clean tank, you probably don't need this, and our app will tell you that.** It's for your neighbours on two hours a day, the households on borewells where all twelve sampled wells had faecal coliform, and the farmer downstream losing a third of his orange yield."

Refusing to oversell here wins the exchange. A judge asking this is testing whether you'll overclaim.

Q7 • "It exceeds a limit. So what do I do?"

"We tell you. And there's one thing almost nobody knows: **boiling fixes bacteria but makes nitrate worse** — it evaporates the water and concentrates the nitrate. **52 of our 347 sampled sites exceed the nitrate limit.** Boiling is the first thing every household does, and for those places it's the wrong move."

Q8 • "Your nearest measurement is 8 km away — that's meaningless."

"Agreed — **and we show you the 8 km instead of hiding it.** Most products would interpolate and hand you a confident colour. We tell you the nearest real test is 8 km away, because that fact *is* the finding."

Q9 • "Does this make the city look bad?"

"It shouldn't. **Nagpur tripled its sewage treatment capacity since 2011** — that's in our material. OCW's own number is 99.6% fit. Our argument isn't that anyone is failing. **The gap is resolution, not effort.**"

Q10 • "Who pays for it after this?"

"Almost nothing to run — public static data, no field hardware. Honestly, **this shouldn't be a startup. It should be a published feed.** If the utility starts publishing per-zone results and we become redundant, we've won."

Q11 • "How do you know your data is right?"

"We extracted it from the source PDFs and **checked it against the reports' own published statistics.** CGWB says 35 of 263 shallow samples exceed nitrate — our extraction returns exactly 35 of 263. Every headline figure reproduces."

Q12 • "Was the sampling seasonal? Coliform spikes post-monsoon."

"We don't know, and it matters." The sampling dates aren't in the tables we extracted, and coliform is seasonally sensitive. So we report it as what that campaign measured and we don't extrapolate.

Answering "we don't know" to a methodology question is stronger than guessing. They're testing whether you'll bluff.

6. THINGS TO NEVER SAY

- ✗ "Our app tells you if your water is **safe**" — we never say safe, and that's a design decision we're proud of
- ✗ "**First**" / "**only**" / "nobody has ever" — unnecessary and attackable
- ✗ "**All** wells in Nagpur are contaminated" — say "*all twelve sampled*"
- ✗ "**Deep borewells are worse**" — the data suggests it but isn't statistically significant
- ✗ "**265 MLD flows untreated**" — that's a 2011 figure. Current: ~520 generated, ~345 treated, ~175 untreated
- ✗ Anything implying the utility is negligent

7. HOW TO CLOSE

"Every organisation here is doing its job. The data exists. It just stops before it reaches the person it's about — and that last step is what we built."

And if there's a moment for an ask:

"What we'd want is a zone-level quality feed. Everything else we can do ourselves."

8. ONE THING TO CARRY IN YOUR HEAD

The whole project is one idea, and everything above is a variation of it:

We are not measuring new things. We are connecting what is already measured to the person it affects, at the resolution they actually live at.

If a question comes that isn't on this sheet, answer it from that sentence. It generalises.